McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

Crary Science and Engineering Center

Location ID: B001

CSEC is a scientific research facility with offices, storage, library, an aquarium, and laboratories for biology, atmospheric science and earth science research. The building is configured in three "phases" at different elevations connected by a walkway spine. A dedicated 1,900 foot pipeline supplies seawater to the building's aquarium from the station seawater intake well.



Photo 1: *Crary Science and Engineering Center*

General Structure Information



Automatic External Defibrillator (AED) Locations

This building contains one or more AED devices that may be used in the event of life-threatening cardiac arrhythmia.

- **Mounted in Phase 1 near Manager's Office**

Usage	Scientific Research laboratory (Albert P. Crary Science and Engineering Center or CSEC)
Occupancy Group	B-2 Office
Maximum Occupancy	285
Construction Type	Type V, 1-hour, combustible; two-story, steel-framed structure with 2-1/2" thick foam-insulated metal sandwich panels on exterior walls, roof, and underfloor soffit, steel-framed plywood floor. Steel-framed gypsum board interior partitions.
Year Constructed	1988 - 1993
Dimensions	Phase I: 231' X 88' Phase II: 242' X 50' Phase III: 63' X 50'
Floor Area	

Floor	Area
Floor 1	37,400 sq ft
Floor 2	8,600 sq ft

Floor	Area
Total	46,000 sq ft

HVAC System

Heating	Two Bryan 322 hp low-pressure steam boilers derated to 150 hp with five glycol heat exchangers with glycol heating loops and VAV zone reheaters. There are two phase-1 heat exchangers, two phase-2 heat exchangers, and 1 phase-3 heat exchanger.
Air Handling Units	Five central-station air handler units
Fuel Tank	25,000 gallon fuel tank

Water and Sewer

Water Service	3" domestic supply 1" seawater supply to Phase I 4" seawater supply to Phase III 1-1/2" fire water supply to Phase I
Sewer Service	4" main stub into 6" line
Connections	Sinks: Toilets:

	Urinals: Showers: Water Coolers:
Water Heater	Four instantaneous steam heat exchangers with two online and two acting as backup

Electrical Service

Service Feeder	Feeder E
Electrical Service	400 kW standby generator (located in Building 1A); 3000A, 120/208V, 3PH main switchboard; 1600A, 120/208V, 3PH emergency main switchboard; 1600A automatic transfer switch.

Fire System

Fire Panel Type	Pyrotronics System 3 control panels (2 panels, 32 and 40 module spaces), King Fisher KFRTI-WP-8 radio interface/transmitter panel (48 zones from system 3 panel consolidated into 8 zones on WP-8), four Pyrotronics remote annunciator panels at entryways, pull stations, and alarm horns.
Heat Detectors	Located throughout buildingLocated in boiler rooms
Suppression Type	Automatic sprinkler system: Grinnell model A-4 mulimatic preaction valves with 24 oz. system air maintenance. Seven 4-inch risers are fed by a pressurized 10,000 gallon storage tank (6,000 gallons of water at 85 psi). Sprinkler opens on heat, causing low air

supervisory.
Preaction valve releases water upon low air supervisory and heat detector in that riser zone by signal
from panel to solenoid valve at riser.

Halon system: Pyrotronics Pyr-A-Lon halon extinguishing system controlled by System 3 control panel.
Heat detector activates alarms and time delay, then Halon release occurs in a small flammables storage room,
which is interfaced with the main control panels.

Additional Photos



Photo 2: *Crary Science and Engineering Center*

Floorplans

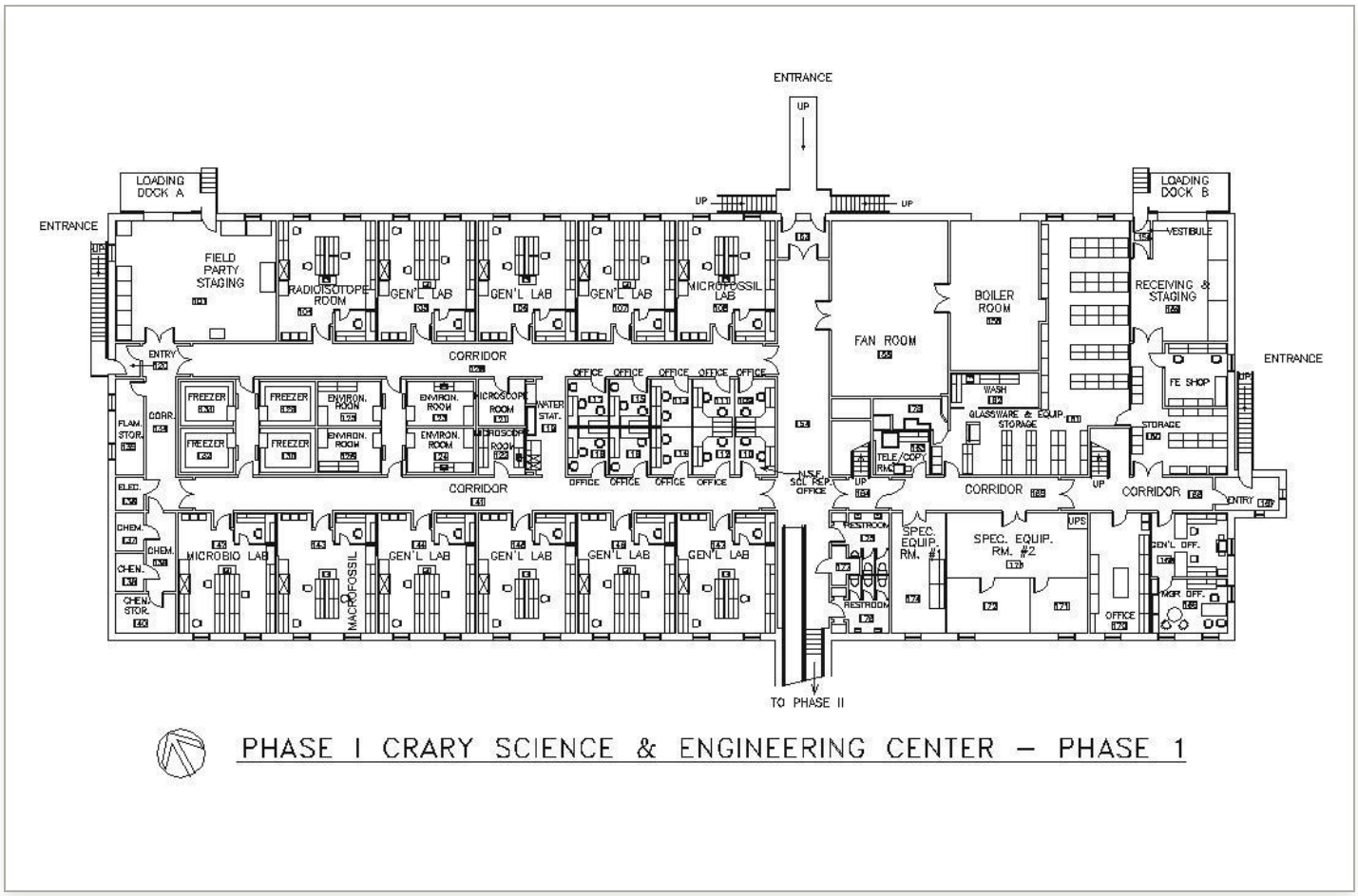


Figure 1: *Crary Science and Engineering Center*

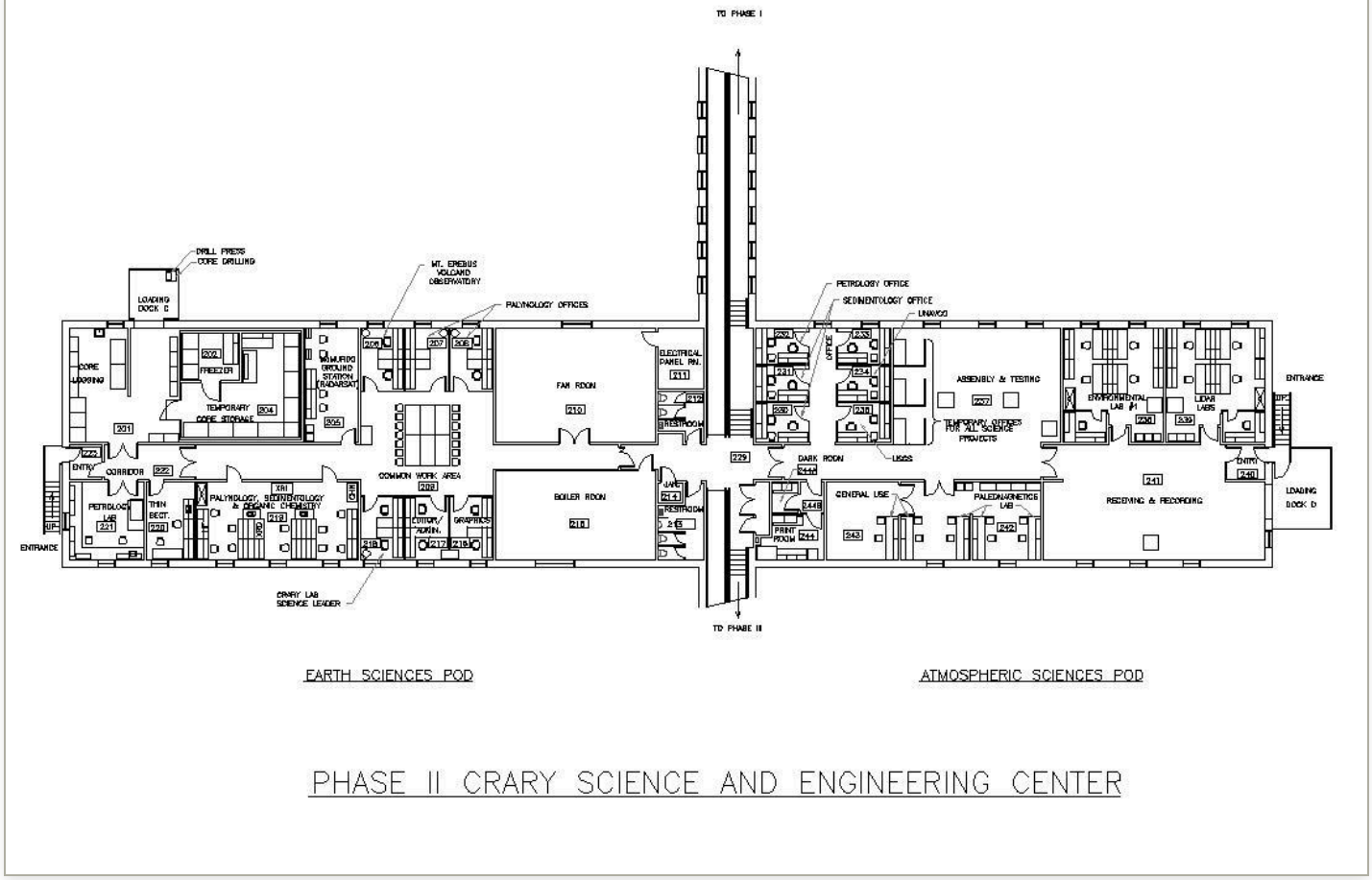


Figure 2: *Crary Science and Engineering Center*

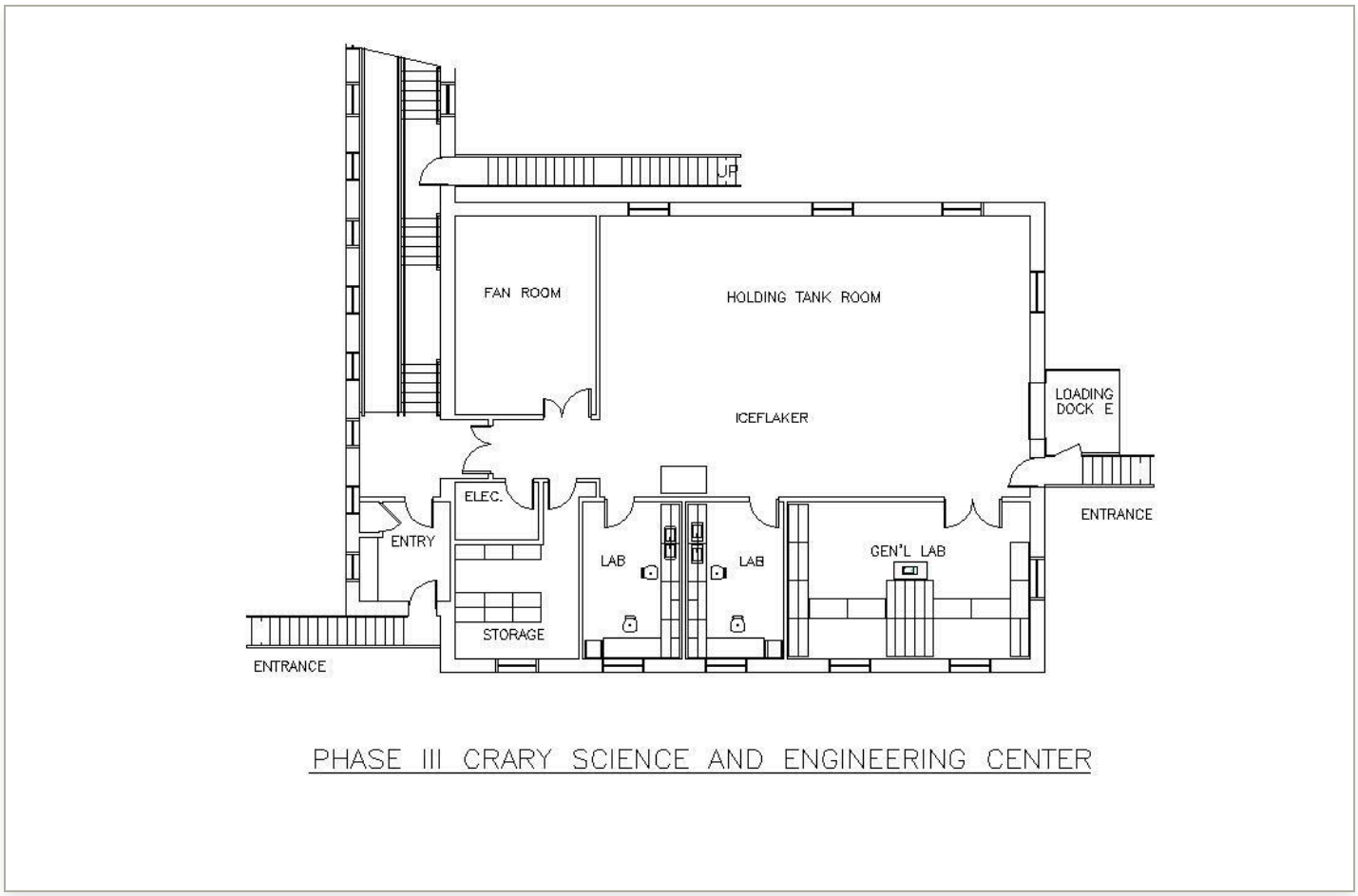


Figure 3: *Crary Science and Engineering Center*